# FULLY FAITHFUL EXACT FUNCTOR WITH ZERO-KERNEL RIGHT ADJOINT IS AN EQUIVALENCE

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*Editorial note.* Counted span is the author's own proof environment, source lines 2603-2618, following the statement at 2589-2601.

*Editorial note.* The two one-step triangulated inferences (identity triangle has zero third object; triangle ending in zero starts with an isomorphism) are expanded via Hom-vanishing readings in the notes.

*Allowed standard prerequisites (editorial).*

- Fully faithful left adjoint has invertible counit-style unit. Categories, Lemma categories-lemma-adjoint-fully-faithful (cross-chapter, cited at ~2605). id -> G o F is an isomorphism.

*Source reference.* Original reference 07RB: focused-seeds/geometry/S9-der-fully-faithful-adjoint-kernel-zero/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $\mathcal{D}$, $\mathcal{D}'$ be triangulated categories. Let $F : \mathcal{D} \to \mathcal{D}'$ and $G : \mathcal{D}' \to \mathcal{D}$ be functors. Assume that*

(1) *$F$ and $G$ are exact functors,*
(2) *$F$ is fully faithful,*
(3) *$G$ is a right adjoint to $F$, and*
(4) *the kernel of $G$ is zero.*

*Then $F$ is an equivalence of categories.*

*Proof.* Since $F$ is fully faithful the adjunction map $\mathrm{id} \to G \circ F$ is an isomorphism (Categories, Lemma 07RB). Let $X$ be an object of $\mathcal{D}'$. Choose a distinguished triangle

$$F(G(X)) \to X \to Y \to F(G(X))[1]$$

in $\mathcal{D}'$. Applying $G$ and using that $G(F(G(X))) = G(X)$ we find a distinguished triangle

$$G(X) \to G(X) \to G(Y) \to G(X)[1]$$

Hence $G(Y) = 0$. Thus $Y = 0$. Thus $F(G(X)) \to X$ is an isomorphism. □

*Date*: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.